Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation

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***Abstract*—Large language models let non-player characters hold conversations, but they forget early facts, propose changes that break game rules, and cost more as context grows. We present ENGINE-X, a dialogue engine that separates generation from state control: a dual-tier memory whose retrieval is scoped to the current location and character, and an event-sourced world state in which every proposed change must pass a symbolic validation barrier before commit. We compare it with six memory baselines at an equal budget, including the retrieval rule of the base paper, Generative Agents, in a memory-by-control factorial with eight ablations, on seven worlds and four models, and audit every committed change with an invariant oracle that shares no code with the validator. With a 7-billion-parameter backbone, it recalled 63.7% of facts 20 or more turns old on held-out worlds, against 0.0% for a rolling context and 54.2% for the base paper's rule, but less than flat vector memory (69.3%) and no more than its earlier version (63.3%). The barrier committed no state violation in probes or live play and accepted none of 26,273 unlawful random proposals, against 4.4% attack success under direct writes. Its cost is grounding: it rejected 20.0% of lawful requests that named entities by display name. Recovery was exact after all 300 simulated crashes.**

***Index Terms*—Large language models, non-player characters, long-term memory, event sourcing, runtime verification.**

I. INTRODUCTION

TEXT-centric games mediate navigation, object interaction and character dialogue through language, so a text world stays consistent only through an explicit symbolic state. Dialogue trees and finite-state machines obey the rules at negligible cost but restrict players to predefined choices. LLMs remove that restriction, and surveys of LLMs in games identify the coupling of generated text to game state as an open problem [1].

Three failure modes stand in the way. First, *long-horizon forgetting*: facts outside a bounded context are lost, facts inside a long context are used unevenly [2], and very long-term conversational memory remains weak even with retrieval [3], [4]. Second, *invariant violations*: LLMs are unreliable simulators of state transitions [5], [6] and lose narrative commitments [7], so unchecked they move players between unconnected rooms, invent items and grant unearned money. Third, *cost growth*: full history, or several reflective LLM calls per turn [8], [9], raises tokens and latency.

We evaluate an engine that separates generation from state control and compare it with the base paper, Generative Agents [8]. Its components (a turn buffer, dense retrieval, schema-constrained output, a rule validator, an append-only event log and a deterministic reducer) are individually standard, so the contribution is an evaluation that isolates what each mechanism buys, against independent baselines, with its costs measured. It asks five questions. **RQ1**: does scoped dual-tier memory improve long-range recall per prompt token over rolling-context, vector-memory and generic retrieval agents? **RQ2**: does a pre-commit validation barrier over event-sourced state reduce committed invariant violations, and at what cost in false rejections and dialogue–state desynchronisation? **RQ3**: do memory and control contribute independently? **RQ4**: what latency and tokens does each component add? **RQ5**: do the effects hold across worlds of different authorship, genre and scale, and across backbones? The contributions are:

1) a formal account of the engine as implemented, including the post-filter limit of its scoped retrieval (Section IV-A),
2) a *paired five-way commit protocol*: each LLM output is committed through the corrected barrier, a canonicalising variant, the earlier barrier, the reducer alone and by direct writes, and audited by an eight-invariant oracle that imports nothing from the validator, with a lawful *twin* for every attack and a differential test on random proposals (Section IV-D),
3) a seven-world benchmark from four families and a single-notebook harness that reports recall, retrieval quality, violations, false rejection, desynchronisation, latency, tokens and replay consistency with pre-registered hypotheses (Section V),
4) findings from the final run of a protocol revised after auditing four earlier runs: a paired before-and-after measurement of the corrections to the first version, the cost of grounding model output in canonical ids, and the hypotheses the corrected system does not support (Section IX).

The supplementary material holds the algorithms, the dataset, the run configuration and audit, and every secondary result. Its sections, tables and figures are numbered with an S prefix.

II. LITERATURE SURVEY

Grounded dialogue and text games. LIGHT conditions dialogue on rooms, objects and personas in a crowd-written fantasy world [10], and goal-driven LIGHT agents were trained with reinforcement learning [11], but neither keeps a validated world state over long sessions. TextWorld [12] and Jericho [13] enforce rules but parse a constrained command language with no open NPC dialogue. Callison-Burch et al. [14] framed Dungeons and Dragons as a dialogue challenge including state prediction, and STORIUM anchors collaborative stories with structured cards [15].

A. Base paper: Generative Agents

Generative Agents [8] is the most direct prior architecture for LLM-driven characters with long-term memory in a game-like world. Twenty-five agents in a sandbox town keep a *memory stream* of natural-language observations. Retrieval scores each memory as the equally weighted sum of min-max-normalised recency (decay 0.995 per sandbox hour), importance (an integer from 1 to 10 that the LLM assigns on creation) and relevance (cosine similarity), and reflections are synthesised when summed importance exceeds 150. Its evaluation ranks believability: 100 human evaluators ranked interview answers of the full architecture and its ablations. Three properties motivate this work. Memory is not bound to a validated symbolic state, and the paper reports embellished memories. Memory is expensive: every write costs an LLM call, and two simulated days of 25 agents cost thousands of dollars in token credits. Finally, neither recall of established facts nor rule violations is measured, at a matched token budget or otherwise.

Memory for LLM agents. Reflexion stores verbal self-reflections [9], and summarising earlier sessions helps multi-session chat [16]. Later systems add forgetting curves [17], knowledge-graph indexing [18], linked notes [19], multi-granularity summaries refined by reinforcement learning [20] and compressed memory units [21]. Benchmarks show that long-term memory is far from solved [3], [4], and surveys organise the design space [22], [23]. None binds memory to a validated symbolic state or compares designs at a matched token budget in a game.

Retrieval and long context. Retrieval-augmented generation conditions generation on passages from a dense index [24] but ranks by similarity alone, with no notion of a turn's spatial or social scope. Longer contexts are no substitute, since models use their middle poorly [2], and ReAct keeps a growing scratchpad [25]. Scoping is a filtered nearest-neighbour query, and post-filtering a global top-k loses in-scope neighbours, which motivates filter-aware indices [26] (Proposition 2).

State tracking and verifier-gated generation. LLMs track entities imperfectly [6] and simulate text worlds unreliably [5]. LLM-Modulo argues for external model-based verifiers of LLM proposals [27], grammar-constrained decoding guarantees syntax but not legality [28], and prompt injection shows that model output cannot be trusted to follow its instructions [29]. Event sourcing derives state from an append-only log in industrial systems that need an auditable history [30].

LLM NPCs and narrative games. PANGeA lets the LLM itself judge player input against design rules [31], and Phillips et al. advance goal-based conversation states with an LLM that reads the dialogue history [32]. SceneCraft generates branching NPC scenes [33], and CALYPSO supports Dungeon Masters with grounded LLM text [34]. NCP-Bench found that the best model kept its narrative commitments in 42% of runs after 20 turns, with fact-conflict rates of 40% to 68%, as judged by LLM auditors [7]. Table I contrasts these systems.

III. Research Gap

The survey leaves five gaps. **RG1:** memory is not bound to validated state (Table I). **RG2:** in-game state changes are judged by a model [31], [32] or not at all, so symbolic pre-commit validation of LLM-proposed mutations is unevaluated in NPC dialogue. **RG3:** violations are scored by the component under test or by LLM auditors that can err on ambiguous text [7], never by a rule checker sharing no code with the validator on identical outputs. **RG4:** memory designs are rarely compared at an equal prompt-token budget, and the base paper reports cost only as token credits. **RG5:** false rejection, dialogue that contradicts rejected changes, and crash recovery are not reported with recall and violations. RQ1 to RQ5 address these gaps.

IV. Proposed Methodology

A. Formal model

The model is the one the code implements. Module paths refer to `Backend/` at the evaluated commit.

Definition 1 (World state)**.** A world state is a tuple $S = (\mathcal{L}, A, \mathcal{N}, \mathcal{O}, \pi, \iota, c, R, J, t)$ where $\mathcal{L}$ is the set of locations with directed adjacency $A \subseteq \mathcal{L} \times \mathcal{L}$, $\mathcal{N}$ the NPCs (location, persona, knowledge, conditional secrets, `alive` flag), and $\mathcal{O}$ the objects (location in $\mathcal{L} \cup \{\perp\}$, owner). The player's location is $\pi \in \mathcal{L}$, the inventory ι and the currency $c \in \mathbb{Z}_{\geq 0}$. The trust map is $R : \mathcal{N} \times (\mathcal{N} \cup \{\text{player}\}) \to [-100, 100]$, the journal J and the turn t (schemas/world_state.py).

The seed S_0 fixes the canonical entity sets $\mathcal{L}_0, \mathcal{N}_0, \mathcal{O}_0$, and the active NPC and player identity are restored from events, so the whole stored state is derived from the log.

1) Events, reduction and replay: An event $e = (id, t, \tau, p, ts)$ has one of 14 types τ (schemas/events.py). The reducer δ (core/reducer.py) deep-copies the state and applies one event, ignoring moves to unknown locations, clamping trust to $[-100, 100]$ and flooring currency at zero, so

$$S_T = \text{foldl}(\delta, S_0, [e_1, \ldots, e_T]), \qquad S_i = \delta(S_{i-1}, e_i). \quad (1)$$

Events are appended to SQLite in one transaction per turn, and every $\Delta = 16$ turns a snapshot (S_k, id_k) is written atomically.

Proposition 1 (Replay and snapshot equivalence)**.** *If (i) δ depends only on (S, e) and (ii) every change to the hashed state is a logged event applied by δ, then for a snapshot after event k,* $\text{foldl}(\delta, S_0, e_{1..T}) = \text{foldl}(\delta, S_k, e_{k+1..T})$.

Proof. A left fold over a concatenated list folds the suffix from the prefix's result, which is S_k. Condition (i) holds because the reducer's only time-dependent value, a journal timestamp, is read from the persisted event. Condition (ii) failed in the Review-02 version for the active NPC, which is now an event field. □

Recovery from the latest snapshot costs $O(|S| + r)$ for the r events after it. The corrected load path (`recover_state`, core/recovery.py) returns $\text{foldl}(\delta, S_k, e_{k+1..T})$, while the Review-02 path returned S_k and discarded every later event.

2) Proposals and the validation barrier: Each turn the LLM returns one JSON object validated by a Pydantic schema: dialogue, narration, a one-sentence memory summary and a list of proposals $P_t = [(\tau_j, p_j)]$. The barrier (game/validator

TABLE I
SURVEYED SYSTEMS AGAINST THE PROPERTIES STUDIED HERE. "INDEP. AUDIT" MEANS VIOLATIONS ARE SCORED BY A CHECKER THAT DOES NOT SHARE CODE WITH THE COMPONENT BEING EVALUATED. ENTRIES DESCRIBE THE CITED PAPERS' OWN EVALUATIONS.

Work	Memory	World state	Pre-commit check	Indep. audit	Replay
LIGHT [10]	episode context	text descriptions	action checks	–	–
D&D dialogue [14]	dialogue history	predicted from text	–	–	–
Generative Agents [8] (base)	scored memory stream	sandbox tree	–	–	–
MemoryBank [17]	forgetting-curve store	none	–	–	–
A-Mem [19]	linked notes	none	–	–	–
TextWorld/Jericho	game engine	formal state	parser and rules	–	engine
PANGeA [31]	memory system	game level data	LLM judge	–	–
Phillips et al. [32]	dialogue history	goal states	LLM judge	–	–
NCP-Bench [7]	model context	fact ledger	–	LLM auditors	–
This work	scoped buffer + FAISS	event-sourced, typed	symbolic validator	yes (8 invariants)	yes

.py) checks them *in order against a running state* $\hat{S}_0 = S_t$, $\hat{S}_j = \delta(\hat{S}_{j-1}, e(\tau_j, p_j))$ when accepted, and commits

$$E_t = [\, e(\tau_j, p_j) \;:\; V(\tau_j, p_j, \hat{S}_{j-1}, S_t)\,], \qquad S_{t+1} = \mathrm{foldl}(\delta, S_t, [e_{\text{speech}}] +\!\!+ E_t). \tag{2}$$

V requires a canonical, adjacent destination and at most one move per turn, a canonical NPC and a total trust change within ± 20 per NPC, a canonical object at the player's location for a take, possession or presence for NPC drops and takes, connected NPC moves, a canonical target for state updates, and integer currency deltas whose total spending stays within the opening balance $c(S_t)$ and whose net gain is at most 50. Engine-only event types and new entities are rejected. After a rejection, a repair node asks the model once to rewrite the reply so that it does not describe the rejected change. The Review-02 barrier, the "before" condition, checked each proposal independently against S_t, did not bound currency gains and had no repair.

3) Invariants audited by the oracle: The evaluation's oracle re-derives facts from $(S_{\text{pre}}, S_{\text{post}}, \text{committed mutations})$ and checks eight invariants written from the specification, not the validator. With $\mathcal{A}$ the undirected closure of the seed adjacency: (O1) entity sets equal the seed's and all references, including trust relationships, are canonical, (O2) each object is in exactly one place, (O3) $\pi_{\text{post}} = \pi_{\text{pre}}$ or $(\pi_{\text{pre}}, \pi_{\text{post}}) \in \mathcal{A}$, (O4) a newly held object was at the player's location, on the floor or held by an NPC there, and is held once, (O5) objects move only to the player's location and NPCs only along $\mathcal{A}$, (O6) $|R_{\text{post}}(n, \text{player}) - R_{\text{pre}}(n, \text{player})| \le 20$, (O7) spending $\le c_{\text{pre}}$ and $c_{\text{post}} \ge 0$, and (O8) $c_{\text{post}} - c_{\text{pre}} \le 50$, a bound the Review-02 validator did not implement.

4) Scoped dual-tier memory: Long-term memory holds entries $m_i = (\mathbf{v}_i, x_i, t_i, \ell_i, n_i)$: the L2-normalised embedding of the player input under all-MiniLM-L12-v2 [35], [36] ($d = 384$), the model's memory summary, the turn, the player's location and the active NPC. An exact inner-product FAISS index [37] returns $\mathrm{top}_m(\mathbf{q})$, the m most similar entries. With the scope predicate

$$\phi_{\ell,n}(m_i) = \big(\ell_i = \varnothing \vee \ell_i = \ell\big) \wedge \big(n_i = \varnothing \vee n_i = n\big), \tag{3}$$

over-fetch $f = 5$, threshold $\theta = 3$ and $k = 6$, retrieval is

$$\mathcal{R}(\mathbf{q}) = \begin{cases} \mathrm{first}_k\big[m \in \mathrm{top}_{\min(fk,|M|)}(\mathbf{q}) : \phi_{\ell,n}(m)\big] \\ \qquad \text{if this list has } \ge \theta \text{ entries}, \\ \mathrm{top}_k(\mathbf{q}) \quad \text{otherwise (fallback)}. \end{cases} \tag{4}$$

The short-term tier is a buffer of the last $B_s = 8$ turns, and the store keeps its newest 500 entries when it passes 1,000. Memories and buffer share one budget of 1,600 characters (llm/prompt_builder.py): the newest two turns first, then memories in rank order, then older turns, and a memory of a turn already in the history is dropped. The prompt also states the validator's limits from the same constants and shows an NPC's secrets once its trust reaches 60.

Proposition 2 (Post-filter reachability)**.** *An in-scope entry can be returned by the scoped branch of* (4) *only if its similarity rank among all entries is at most* $fk = 30$*, so scoped recall falls below exact filtered search when more than* $fk - 1$ *out-of-scope entries are more similar.*

Proof. The scoped branch inspects only $\mathrm{top}_{fk}(\mathbf{q})$. □

B. Turn pipeline

Fig. 1 shows one turn. Only accepted proposals are appended to the event log and applied by the reducer, and the commit node also updates the buffer, the FAISS store and, every 16 turns, the snapshot.

C. Experimental conditions

All conditions share the backbone, decoding, system prompt, state serialisation, output schema and player scripts, and differ only in memory and in how proposals reach state (Table II). Baselines are independent NumPy implementations, and every memory condition except B0, F4v1 and A8 is held to 1,600 characters. Budget B for the rolling contexts is the mean token count of F4's memory and history, 401 tokens. Without a validator, schema-valid proposals are written straight into the state (direct writes). B4 re-implements the base paper's retrieval rule, with importance rated once per memory by the primary model, and B5 MemoryBank's forgetting curve [17]. F4v1 is the Review-02 system on the corrected reducer, event store and commit node.

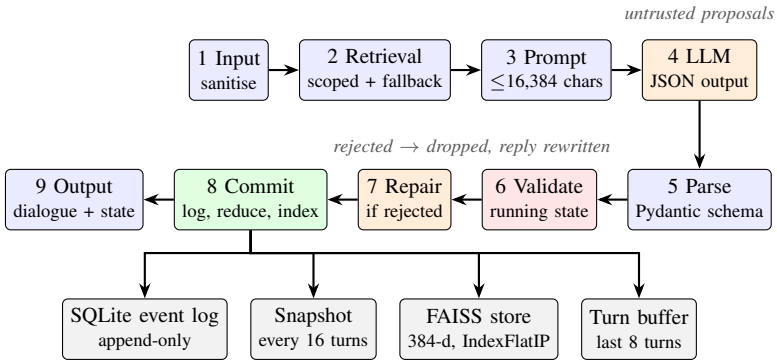


Fig. 1. Turn pipeline of ENGINE-X (graph/definition.py). The LLM output is untrusted: proposals reach the world state only through the validation barrier (node 6) and the pure reducer inside the commit node (node 8). Node 7 rewrites the reply when a proposal was rejected. The commit node writes the four stores. On later turns, retrieval (node 2) reads the FAISS store and prompt assembly (node 3) packs the memories and the turn buffer into one budget.

TABLE II
CONDITIONS. F1 TO F4 FORM THE 2×2 FACTORIAL. A1 TO A8 CHANGE ONE COMPONENT OF F4.

ID	Long-term memory	Prompt history	State control
F4	real `FAISSMemory` + real retrieval node (scope, fallback), packed into the budget	8-turn buffer, in the same budget	barrier + repair + event log + reducer (as shipped)
F4v1	as F4, unbudgeted, Review-02 prompt	8-turn buffer	Review-02 barrier, no repair
B0	none	full history, oldest dropped only at the context limit	direct writes
B1	none	newest turns within B tokens	direct writes
B2	flat store of turn *summaries*, cosine rank, no scope	none	direct writes
B3	raw dialogue chunks + world lore documents, cosine rank	none	direct writes
B4	scored memory stream: relevance + recency (decay 0.995 per turn) + importance	none	direct writes
B5	forgetting curve: relevance $\times \exp(-\Delta t/(S\tau))$, $\tau = 50$	none	direct writes
F1	none	rolling, budget B	direct writes
F2	none	rolling, budget B	barrier + repair + event log
F3	as F4	8-turn buffer	direct writes
A1	none	8-turn buffer	as F4
A2	as F4	none	as F4
A3	unscoped top-6	8-turn buffer	as F4
A4	scoped, no fallback ($\theta = 0$)	8-turn buffer	as F4
A5	as F4	8-turn buffer	schema-only events through the reducer
A6	as F4	8-turn buffer	validator, then in-place writes, no log
A7	as F4	8-turn buffer	as F4, recovery by full replay
A8	as F4, unbudgeted (all six memories)	full 8-turn buffer	as F4

D. Paired five-way commit protocol

Request seeds depend on the probe or turn, never on the condition, so identical prompts receive identical outputs. A probe prompt carries three history turns, which every memory condition shows in full, so probes are posed once, under F4's context. Each parsed output is committed five ways (corrected barrier, the barrier with a canonicaliser that maps display names to ids, Review-02 barrier, reducer alone, direct writes) and audited by the oracle, so differences between modes are due to the commit path alone. Lawfulness is judged by the oracle on a single-proposal transition, both as written and *by intent*, with display names and slugs mapped to ids, so false-rejection and validator-gap rates do not depend on the validator's verdicts (Algorithm S3). A differential test drives the barriers directly with random proposals of every event type in random valid states, labelled by the oracle.

E. Metrics

Recall is strict: a probe is correct when the output parses, the reply names the gold key word, and it names neither the distractor's value nor, after a knowledge update, the outdated one. The earlier lenient rule and an adjudication of hedged replies by another model family are reported as secondary analyses. With r_j the rank of probe j's gold memory, $\mathrm{MRR@}k = \frac{1}{n}\sum_j \mathbb{1}[r_j \le k]\, r_j^{-1}$ and $\mathrm{Recall@}k = \frac{1}{n}\sum_j \mathbb{1}[r_j \le k]$, computed on probes older than the 8-turn buffer, because packing drops a memory whose turn is buffered. *Attack success* is the share of attacks whose committed changes violate an invariant (a secret leak for secret extraction), *false rejection* the share of lawful twin proposals rejected, and the *validator gap* the share of unlawful proposals accepted. The *violation rate* is 100 times the number of turns whose own committed changes violate an invariant over the committed non-speech mutations, and the share of turns that end in a corrupted state is reported beside it. *Desynchronisation* flags a non-move turn whose dialogue lacks a refusal cue after a rejection, or claims an arrival the state did not commit. Two *contradiction judges* of other model families (Qwen2.5-7B-Instruct and Phi-4-mini-instruct) label replies that claim uncommitted changes, and a judge's verdicts are used only if it labels at least 80% of 27 control items correctly. *Replay consistency* compares SHA-256 hashes of the rebuilt and live states, and a *crash trial* stops a commit at a random point and compares the recovered state with the last committed one.

F. Statistical analysis

Intervals are 95% cluster-bootstrap percentile intervals with 10,000 resamples [38], [39], clustered on session for recall and live metrics and on attack template for pooled probe metrics, with ratios bootstrapped as ratios of resampled sums. A proportion with no event, or with every row an event, gets an exact Clopper–Pearson interval [40]. Paired binary outcomes use the exact McNemar test [41] and paired continuous outcomes the Wilcoxon signed-rank test [42]. The six confirmatory tests of H1 to H3 are Holm-corrected [43]. The factorial is modelled on live turns as a linear probability model

$$\Pr(y = 1) = \beta_0 + \beta_M M + \beta_C C + \beta_{MC} MC, \qquad (5)$$

fitted in statsmodels [44] with standard errors clustered on the 14 live scripts and a t reference with $G - 1 = 13$ degrees of freedom, as recommended for few clusters [45]. Agreement between judges is Cohen's κ [46].

G. Hypotheses

Stated before the run, each against the strongest applicable baseline and supported only if the corrected effect is significant with the predicted sign or its threshold is met. **H1a**: F4 recall at $a \ge 20$ exceeds B1 and B3 at equal or lower token cost. **H1b**: it is non-inferior to B2 within 5 points. **H2**: scoping plus fallback gives higher MRR@6 than B2. **H3**: the barrier reduces committed violations on probes and in live sessions. **H4**: false rejection by intent is at most 5% and validation adds under 5 ms at P95. **H5**: replay reproduces the live state in every event-sourced session. **H6**: memory affects recall and control affects violations additively, without significant interaction. **H7**: the H1 and H3 directions hold on W2, W3 and W4 and on two other model families. Four tests compare the system with its Review-02 version: **R1** currency, **R2** crash recovery, **R3** contradictions and **R4** live violations.

V. Dataset Description and Usage

The benchmark is generated deterministically by the evaluation notebook from a master seed (Section S2). It has seven worlds in four families (Table S1): W1, the in-house development world of 8 locations, W2a to W2c (16 to 32 locations) converted from the crowd-written LIGHT environment [10] (CC BY-NC 4.0), W3, a 24-location science-fiction station written in the notebook, and the procedural W4-32 and W4-128. No setting was tuned on W2 to W4. *Recall sessions* are teacher-forced scripts of 50, 100 or 200 turns in which facts are planted with an NPC and probed 5 to 160 turns later, each with a paraphrased distractor told to another NPC, so every condition sees the same history. A quarter of facts are corrected later (knowledge updates), half have a topical lure turn, and each attribute is planted at most once per session. Each condition answers 1,045 probes, 897 on held-out worlds, of which 537 have $a \ge 20$. *Adversarial probes* cover seven attack categories (teleportation, fabrication, theft, trust inflation, currency, prompt injection [29] and secret extraction) with six templates each, giving 1,078 attacks and 1,027 lawful twins. *Live sessions* run the real backend graph for 39 turns with plants, walks, trades, six attacks and four late probes, giving 14 sessions per condition. The data, predictions, metrics and blinded annotation sheets are released with the code.

VI. Novelty

We claim no novelty for the engine's components. Against the base paper, the engine scopes retrieval by location and NPC with no LLM call per memory write, binds every model-proposed change to a typed, event-sourced state through a symbolic barrier and a pure reducer, and is evaluated by automatic, paired, oracle-audited metrics at a matched token budget instead of believability rankings. To our knowledge, no earlier evaluation pairs symbolic pre-commit validation with an independent oracle on identical outputs. It yields three new

findings. (i) Of the 4.4 points of attack success under direct writes, the reducer's own semantics remove 2.4, the Review-02 validator 1.0 more and the corrected validator 0.8, leaving two secret leaks in dialogue that no state check can block, so validator and reducer must be evaluated as a pair. (ii) An oracle written from the specification exposed a currency rule the first validator never implemented (a 47.6% validator gap on currency proposals) and confirms its repair on 26,273 unlawful random proposals. (iii) The costs of control are measured, and the largest is grounding: the barrier rejects lawful requests that name entities by display name instead of id, which a canonicaliser removes offline.

VII. Algorithms

Three algorithms define the system and its evaluation. Algorithm S1 is one turn of the engine, following nodes 1 to 9 of Fig. 1, Algorithm S2 is the retrieval of (4), and Algorithm S3 is the paired five-way commit and audit. With $|M| \leq 1000$ entries and p proposals per turn, exact search is $O(|M|d)$, validation $O(p)$ lookups plus one reduction per accepted proposal, reduction $O(p\,|S|)$, and recovery $O(|S| + r)$.

VIII. Implementation

The engine has four tiers: a React and TypeScript client over a WebSocket, a FastAPI service with per-session locks running a compiled nine-node LangGraph `StateGraph` (graph/definition.py), the barrier, and storage (the SQLite event log and snapshots, the FAISS store and the buffer). The shipped game uses a hosted model (Groq, `openai/gpt-oss-120b`, temperature 0.35), and the evaluation self-hosts models with vLLM [47]. The evaluation notebook notebooks/npc-memory-state-benchmark.ipynb imports the real backend modules of the evaluated commit from a hashed bundle, so only the baselines, the oracle and the generators are new code. The reported run used two Tesla T4 GPUs on Kaggle for 9.66 hours, `IFM/K2-Horizon-7B` as the primary backbone, and level 5 of twelve pre-declared sample sizes, reached by extending a planned level 3 while time remained (Tables S2 and S19). Audits of four earlier runs found and fixed measurement defects in the notebook (Section S3-E).

IX. Comparison of Results

Values are mean [95% CI], pooled over the held-out worlds W2a to W4-128 unless stated, with K2-Horizon-7B as backbone.

A. Main comparison

Table III summarises the comparison. At the same budget, the budgeted conditions used 1,455 to 1,516 input tokens per turn. F4's 63.7% is significantly below flat vector memory B2 (69.3%, Holm $p = 0.044$ in the exploratory family), not different from F4v1 (63.3%) or B5 (60.7%), and significantly above the scored memory stream B4 (54.2%, $p = 0.0011$), generic RAG (38.0%) and the rolling context (0.0%) (Section IX-G). Full history recalled 53.4% and cost three times F4's input tokens for lower recall.

By fact age (Fig. 2), the rolling context falls to zero once a fact leaves its window, F4, F4v1, B2 and B5 stay roughly flat from age 20 to 160, and B4 falls to 33.3% at age 80 and 15.0% at age 160 as recency pushes old memories out of the budget. F4 is weakest at age 5 (30.0%, against 71.1% for B2 and 74.4% without the buffer, A2). There the fact and its distractor both sit in the buffer, and F4 named both values in 41.1% of replies, which strict scoring counts as wrong. Knowledge updates were hard for every memory: F4 gave the corrected value for 20.7% of updated facts and the outdated one for 4.3% (Section S4-B).

B. Retrieval quality

Scoping changed *ranking* more than *reach*. On the 717 held-out probes older than the buffer, F4, B2, A2 and A4 retrieved the gold memory for at least 99.6% of probes, and F4 ranked it slightly higher than B2 (MRR@6 0.805 against 0.786, Holm $p = 0.065$) and higher than the unscoped A3 (0.742). F4 kept the distractor out of 44.6% of prompts, B2 out of 0.9%. Disabling the fallback (A4) raised MRR@6 to 0.815 (sweeps and seeds in Section S4-B).

C. State control under adversarial probes

Table IV commits each probe output five ways. The corrected barrier accepted no unlawful proposal (validator gap 0.0%, against 12.7% for the Review-02 validator and 47.6% of its currency proposals), and its only attack successes were two secret leaks in dialogue. Much of the protection comes from the reducer: committing every schema-valid proposal as an event (A5) cut attack success from 4.4% to 2.0%, and its remaining successes are 19 currency exploits, one trust jump and the two leaks. The differential test confirms the barrier at scale (Table S5): of 26,273 unlawful random proposals it accepted none, against 13,587 (51.7%) for the Review-02 validator, and no turn of 16,800 committed through it violated an invariant, against 1,485 for Review-02. It rejected 6 of 7,382 lawful random proposals.

The cost of grounding. Strict false rejection was 0.0% on lawful twins, but the model often named an NPC, object or place by its display name instead of its id. Judged by intent, the barrier rejected 17.9% of 408 lawful-by-intent probe proposals and 20.0% of lawful twin proposals, and in live play 63.7% of F4's lawful-by-intent proposals (51 of 80). A canonicaliser that maps exact display names and slugs to ids before validation, evaluated offline on the same outputs, cut the probe rate to 0.0 [0.0, 0.9]% with no validator gap and no added attack success (Table S6).

D. Live factorial, ablations and recovery

In live sessions (Table S7) the model committed only 2.9% to 4.9% of scripted moves itself, so the engine completed the rest. Direct writes committed a new violation on 80 of F1's 103 and 80 of F3's 101 transitions, mostly non-canonical entities (O1) and non-adjacent moves (O3). Neither corrected-barrier condition had one, and F4v1 had six currency gains (O8). Live recall was 46.4% (F3) and 48.2% (F4), although the plant's

TABLE III

MAIN COMPARISON ON HELD-OUT WORLDS. VIOLATIONS AND ATTACK SUCCESS ARE FROM THE PROBE SUITE, WHOSE OUTPUTS ARE THE SAME UNDER EVERY MEMORY CONDITION: F4 COMMITS THEM THROUGH THE CORRECTED BARRIER, F4V1 THROUGH THE REVIEW-02 BARRIER AND THE BASELINES BY DIRECT WRITES (ONE VALUE, IN THE B1 ROW). MRR@6 IS SCORED ON PROBES OLDER THAN THE BUFFER (SECTION IV-E). LATENCY IS SINGLE-STREAM LLM TIME PER REQUEST.

Condition	Recall, $a \geq 20$ (%)	Viol./100 trans.	Attack succ. (%)	MRR@6	P50/P95 LLM (ms)	Input tok.
B0 Full history	53.4 [49.0, 58.1]	as B1	as B1	–	10,737 / 18,238	4,438
B1 Rolling, B	0.0 [0.0, 0.7]	26.5 [19.0, 35.7]	2.6 [1.2, 4.2]	–	5,922 / 7,766	1,513
B2 Vector memory	69.3 [65.5, 73.2]	as B1	as B1	0.786 [0.767, 0.804]	6,269 / 8,139	1,456
B3 Generic RAG	38.0 [34.4, 41.7]	as B1	as B1	0.726 [0.703, 0.748]	5,790 / 11,501	1,516
B4 Memory stream (GA)	54.2 [50.1, 58.4]	as B1	as B1	0.346 [0.321, 0.370]	6,814 / 8,881	1,460
B5 Forgetting curve	60.7 [56.6, 65.0]	as B1	as B1	0.788 [0.766, 0.810]	6,542 / 7,723	1,455
F4v1 Review-02 system	63.3 [59.6, 67.1]	1.6 [0.0, 4.8]	0.5 [0.0, 1.3]	0.751 [0.732, 0.772]	6,452 / 7,478	1,425
F4 Proposed	**63.7 [59.4, 68.0]**	**0.0 [0.0, 1.5]**	**0.1 [0.0, 0.3]**	**0.805 [0.787, 0.824]**	6,750 / 7,847	1,468

Replay consistency: F4 100.0 [76.8, 100.0]% (14 live sessions). The baselines keep no event log. Latency: 12 single-stream recall requests per condition. F4's attack successes are secret leaks in dialogue.

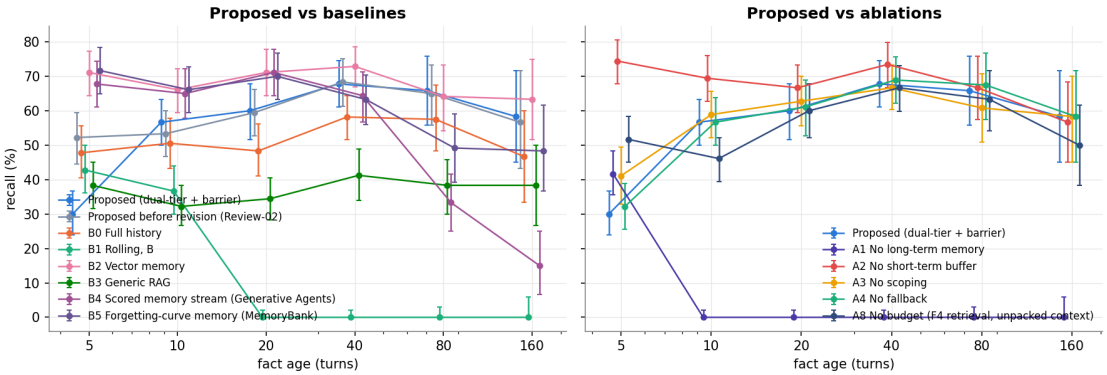


Fig. 2. Strict recall by fact age on held-out worlds with 95% bootstrap intervals. Left: F4 against the baselines and F4v1. The rolling context falls to zero once a fact leaves the window, B4 declines after age 40, and F4 is lowest at age 5, where the distractor shares the buffer with the fact. Right: F4 against its memory ablations. Only removing long-term memory (A1) is catastrophic, and removing the buffer (A2) helps at short ages.

TABLE IV

ADVERSARIAL PROBES BY CATEGORY, ALL WORLDS, F4 CONTEXT (154 ATTACKS PER CATEGORY, 1,078 IN ALL). ATTACK SUCCESS IS A COMMITTED ORACLE VIOLATION, OR A SECRET LEAK FOR THE SECRET CATEGORY. VALIDATOR GAP: UNLAWFUL PROPOSALS ACCEPTED OVER UNLAWFUL PROPOSALS. THE CANONICALISING BARRIER GAVE THE SAME ATTACK SUCCESS AS THE CORRECTED BARRIER IN EVERY CATEGORY.

	Attack success (%)					Validator gap (%)	
Category	Direct writes	Reducer only (A5)	Review-02 barrier	Corrected barrier	False rejection (%)	Review-02	Corrected
Teleportation	3.2 [0.6, 6.5]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 11.2]	4.0 [0.0, 10.2]	0.0 [0.0, 7.1]
Fabrication	1.9 [0.0, 4.5]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 3.4]	4.5 [0.0, 15.4]	0.0 [0.0, 15.4]
Theft	9.7 [5.2, 14.9]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 5.2]	4.3 [0.0, 15.4]	0.0 [0.0, 14.8]
Trust inflation	0.6 [0.0, 1.9]	0.6 [0.0, 1.9]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 41.0]	0.0 [0.0, 97.5]	0.0 [0.0, 97.5]
Currency	13.0 [7.8, 18.2]	12.3 [7.1, 17.5]	5.8 [2.6, 9.7]	0.0 [0.0, 2.4]	0.0 [0.0, 10.9]	47.6 [25.0, 71.4]	0.0 [0.0, 16.1]
Injection	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	0.0 [0.0, 2.4]	no lawful props.	–	–
Secret	1.9 [0.0, 4.5]	1.3 [0.0, 3.2]	1.3 [0.0, 3.2]	1.3 [0.0, 3.2]	0.0 [0.0, 18.5]	100.0 [100.0, 100.0]	0.0 [0.0, 97.5]
All	4.4 [2.3, 6.8]	2.0 [0.4, 4.2]	1.0 [0.1, 2.7]	0.2 [0.0, 0.5]	0.0 [0.0, 1.4]	12.7 [2.6, 28.2]	0.0 [0.0, 3.1]

Most frequent oracle codes under direct writes: theft 14×O1, currency 10×O7 and 9×O8, teleportation 5×O1 and 3×O3. Parse success: 94.0%. Secret leaks over all attacks: 1.0 [0.5, 1.7]%, an upper bound (Section XI).

memory was retrieved for every live probe: in 19.6% of F4's probes the model's own summary had lost the value (Table S8). The linear probability model (Table S9) separates the factors: memory → recall $\beta_M = 0.464$ ($p = 8.6 \times 10^{-5}$), control → violation $\beta_C = -0.147$ ($p = 2.1 \times 10^{-7}$), and β_{MC} with $p = 0.75$ and 1.0. Among the ablations (Table S10), only removing long-term memory (A1) was catastrophic (0.0%). Removing the buffer (A2, 67.8%), the fallback (A4, 64.8%), scoping (A3, 63.1%) or the budget (A8, 61.8%) did not lower recall significantly, and removing the validator (A5) let violations through on probes (4.4 per 100 transitions) and in live play (1.9). Every event-sourced session replayed exactly, snapshot loading took 1.1 to 1.5 ms against 244.2 ms for full replay at turn 160, and the corrected load path was consistent in all 300 crash trials, the Review-02 path in 93 (Table S11).

E. Before and after the corrections

Table V pairs the two systems on the same scenarios. R1 and R2 are confirmed. Only the Qwen2.5-7B judge passed its control items (81%, Phi-4-mini 56%, $\kappa = 0.22$ between them), so R3 uses its verdicts alone. Repair rewrote every rejected reply (117 probe outputs, 61 live F4 turns), yet the judge labelled about as many repaired probe replies contradictory as drafts. F4 contradicted rejected changes on 8.0% of judged live turns against 38.9% for F4v1, but only 7 sessions had judged turns in both, so R3 is not confirmed, and neither is R4 (6 live violations against none).

TABLE V
BEFORE (REVIEW-02 SYSTEM) AND AFTER (CORRECTED SYSTEM) ON THE SAME SCENARIOS, SCORED BY THE SAME ORACLE AND JUDGE (POINT ESTIMATES, INTERVALS IN METRICS/TABLE8_BEFORE_AFTER.CSV). p IS HOLM-CORRECTED OVER THE PAIRED TESTS.

	Measure	Before	After	n	p_{Holm}
R1	attack success through the barrier	1.0%	0.2%	1,078	0.031
	attack success, currency	5.8%	0.0%	154	0.031
	validator gap, currency proposals	47.6%	0.0%	–	–
	false rejection of lawful twins	0.0%	0.0%	–	–
R2	crash trials recovered inconsistently	69.0%	0.0%	300	8.8×10^{-62}
R3	judged contradictions, probe replies, draft vs repaired	48.7%	47.0%	117	1.0
	judged contradictions, live F4 turns, draft vs repaired	10.0%	5.0%	20	1.0
	judged contradictions per live turn, F4v1 vs F4	38.9%	8.0%	7	0.31
	desync proxy per live turn, F4v1 vs F4	4.1%	3.6%	14	1.0
R4	committed live violations per 100 transitions	13.0	0.0	14	0.31

F. Latency and generalisation

Local work took 15.0 ms at P50 and 22.4 ms at P95 for the live F4 graph, and validation 0.96 ms at P95, against a median LLM call of 7,410 ms (Table S17). Per 100 turns F4 used 146,765 input tokens against 443,762 for full history. The recall effect over B1 held on every world family (+0.636 on W2, +0.689 on W3, +0.612 on W4) and for all four models, and the barrier lowered attack success on every family (+0.032, +0.019, +0.016, Table S12). Models of other families proposed far more unlawful changes: direct writes of their outputs succeeded in 49.5% (Qwen2.5-7B) and 27.3% (Phi-4-mini) of attacks on the backbone subset, against 4.4% for K2-Horizon-7B on all worlds, and the corrected barrier let none through for any model.

G. Hypothesis tests

Table VI summarises the verdicts. H1a holds on recall, but F4 used 12 more input tokens per turn than B2, so its token condition is not met. H1b fails (lower bound −9.1 points, −9.8 under adjudicated scoring), H2 is not significant, and H4 fails on false rejection by intent.

TABLE VI
PRE-REGISTERED HYPOTHESES. EFFECTS ARE DIFFERENCES IN PROPORTIONS (MCNEMAR) OR MEAN PAIRED DIFFERENCES (WILCOXON). p IS HOLM-CORRECTED OVER THE SIX CONFIRMATORY TESTS. S: SUPPORTED, P: PARTLY, N: NOT SUPPORTED.

H	Test	n	Effect	p_{Holm}	
H1a	recall $a \geq 20$, F4 vs B1 (McNemar)	537	+0.637	1.3×10^{-102}	P
	recall $a \geq 20$, F4 vs B3	537	+0.257	7.6×10^{-21}	
H1b	F4 vs B2, non-inferiority 5 points	537	−0.056	0.61	N
H2	MRR@6, F4 vs B2 (Wilcoxon)	717	+0.019	0.065	N
H3	attacks, direct vs barrier (McNemar)	1,078	+0.042	2.3×10^{-13}	S
	live violations, F3 vs F4 (Wilcoxon)	14	+0.770	0.0028	
H4	false rejection by intent, validation P95	–	20.0%, 0.96 ms	–	N
H5	replay reproduces live state	42+live	100%	–	S
H6	interaction, recall and violation	14	$p = 0.75$, 1.0	–	S
H7	H1, H3 directions on all families	–	all positive	–	S

H. Comparison with the base paper

The evaluations use different worlds, models and metrics, so the comparison is architectural (Table S13). Memory is necessary in both: removing it lowered the base paper's TrueSkill rating from 29.89 to 21.21, and recall here from 63.7% to 0.0% (A1). The base paper's retrieval rule forgets under a budget (B4), and state control has no counterpart there. F4 had no retrieval miss: its failures were hedged replies, retrieved gold turns it ignored, and confusions (Table S18).

X. DISCUSSION

Memory. At an equal budget, flat vector memory recalled more than scoping and the forgetting curve about as much, and scoping's ranking benefit over flat memory was small and not significant, on a benchmark that favours it (facts are probed with the NPC who heard them). The buffer cost recall at short ages because it carries the distractor too, and the model hedges between values more than it forgets them. At scale, post-filtering the global top-30 can drop in-scope memories that a filtered index keeps (Proposition 2, [26]).

State control. The barrier is only as good as its rule set: the oracle found a missing currency rule, and with it and per-turn checking no violation got through for any model or any random proposal. Its main cost is grounding. Models refer to entities by name, and an id-only validator rejects those requests, so a canonicaliser belongs in front of the barrier.

Blocking still leaves the narration wrong on the turn itself, and one repair call did not fix it. Practitioners should treat LLM output as untrusted input to a state machine, canonicalise references before a per-turn validator, audit with a checker that shares no validator code, and recover by snapshot plus suffix.

XI. Limitations and Threats to Validity

The run used level 5 of twelve sample sizes, with 14 live sessions per condition, and the other backbones ran at level 2 on four worlds. Scoring is automatic, the one contradiction judge that passed its controls is an LLM, and the blinded sheets for two human raters are not yet filled. The leak scorer and the desynchronisation proxy are keyword approximations. Intent-level false rejection is a lower bound (exact names only), and the canonicaliser ran offline only. W3 was not independently authored, players are scripted, the engine completed scripted moves in live play, the models have at most 7B parameters, and the game's hosted model was not evaluated. B4 and B5 re-implement retrieval rules, not full systems, and F4v1 changes validator, prompt, packing and repair together, of which A8 separates only packing from the prompt for recall. Claims are restricted to the tested worlds and models.

XII. Conclusion

We tested an engine that pairs scoped dual-tier memory with validated, event-sourced state, against independent baselines at an equal budget and against its own earlier version. Four claims are supported: long-term retrieval keeps recall flat over 160 turns where an equal rolling context forgets, the barrier committed no state violation for any model or random proposal, memory and control act independently, and snapshot-plus-suffix recovery survived every simulated crash. Four are not: scoped memory does not match flat vector memory, it recalls no more than its earlier version, display-name grounding makes the barrier reject a fifth of lawful requests, and one rewrite of a rejected reply did not reduce contradictions. Future work should canonicalise references in the shipped engine, collect human labels, generate narration from committed events, and probe recall away from where a fact was told. The W2 worlds derive from LIGHT under CC BY-NC 4.0, and the K2-Horizon models are Apache 2.0.

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